BIO 201 - Lecture 7
Cellular Respiration: Harvesting Chemical Energy
Dr. Amara Okafor - Fall 2026

Learning objectives for today: (1) trace the carbon in glucose through glycolysis, pyruvate oxidation and the citric acid cycle; (2) explain how the electron transport chain builds a proton gradient; (3) account for where ATP is made and how much; (4) compare aerobic respiration with fermentation.

The overall reaction is C6H12O6 + 6 O2 -> 6 CO2 + 6 H2O + energy. Roughly 34 percent of the energy in glucose ends up in ATP; the rest is released as heat, which is why exercise warms you up.

A note on numbers: in this course we use the modern estimate of about 30 to 32 ATP per glucose, not the older textbook value of 36 to 38. The older figure assumed whole-number ratios for the proton pump and ignored the cost of moving ATP and phosphate across the mitochondrial membrane. If an exam question asks for total yield, 30-32 is the answer we will accept.

Respiration happens in four stages: glycolysis in the cytoplasm, pyruvate oxidation and the citric acid cycle in the mitochondrial matrix, and oxidative phosphorylation on the inner mitochondrial membrane.

Stage 1: Glycolysis

Glycolysis is a ten-step pathway that runs in the cytoplasm and does not require oxygen. It splits one six-carbon glucose into two three-carbon pyruvate molecules.

The first half is the energy investment phase: the cell spends 2 ATP to phosphorylate glucose and fructose-6-phosphate, which traps the sugar in the cell and destabilizes it so it can be split. The second half is the energy payoff phase, which produces 4 ATP by substrate-level phosphorylation and 2 NADH. The net yield of glycolysis is therefore 2 ATP, 2 NADH and 2 pyruvate per glucose.

Regulation: the key control point of glycolysis is phosphofructokinase-1 (PFK-1), which catalyzes step 3. PFK-1 is allosterically inhibited by high levels of ATP and by citrate, and it is activated by AMP. In plain terms, when the cell already has plenty of energy, glycolysis slows down; when AMP builds up because ATP is being used, glycolysis speeds up.

Exam tip from Dr. Okafor: know that PFK-1 is the main control point of glycolysis and be able to name one inhibitor and one activator.

Stage 2: Pyruvate Oxidation

Pyruvate made in the cytoplasm is actively transported into the mitochondrial matrix. There the pyruvate dehydrogenase complex (PDH) removes one carbon as CO_2, oxidizes the remaining two-carbon fragment, and attaches it to coenzyme A, producing acetyl-CoA.

Per pyruvate this step yields 1 NADH and 1 CO_2, so per glucose it yields 2 NADH and 2 CO_2. This is the first point at which carbon from glucose leaves as carbon dioxide.

Clinical connection: PDH needs thiamine pyrophosphate as a cofactor. Thiamine (vitamin B1) deficiency, called beriberi, impairs PDH, so pyruvate backs up and is converted to lactate. Patients can show lactic acidosis and neurological symptoms because neurons rely so heavily on glucose oxidation.

Remember: pyruvate oxidation is not technically part of the citric acid cycle. It is the link reaction that connects glycolysis to the cycle.

Stage 3: The Citric Acid Cycle (Krebs Cycle)

The citric acid cycle runs in the mitochondrial matrix. It begins when acetyl-CoA (2 carbons) combines with oxaloacetate (4 carbons) to form citrate (6 carbons), a reaction catalyzed by citrate synthase. Over the next seven steps two carbons are released as CO_2 and oxaloacetate is regenerated, which is why it is a cycle.

Each turn of the cycle produces 3 NADH, 1 $FADH_2$, 1 GTP (which is converted to ATP) and 2 CO_2. Because each glucose produces two acetyl-CoA, the cycle turns twice per glucose, giving 6 NADH, 2 $FADH_2$, 2 ATP and 4 CO_2.

Succinate dehydrogenase is the only citric acid cycle enzyme embedded in the inner mitochondrial membrane rather than dissolved in the matrix. It is the same protein as Complex II of the electron transport chain, which is why $FADH_2$ electrons enter the chain at Complex II.

By the end of the cycle all six carbons of the original glucose have been released as CO_2, but most of the energy is now stored in NADH and $FADH_2$, not in ATP.

Stage 4: Oxidative Phosphorylation

The electron transport chain is a series of four protein complexes (I-IV) in the inner mitochondrial membrane. NADH donates electrons at Complex I; FADH2 donates at Complex II. Electrons pass to ubiquinone, Complex III, cytochrome c and finally Complex IV, where oxygen is the final electron acceptor and combines with electrons and protons to form water.

As electrons move down the chain, Complexes I, III and IV pump protons from the matrix into the intermembrane space. The resulting proton gradient (the proton-motive force) drives protons back through ATP synthase, which phosphorylates ADP. This coupling is called chemiosmosis. Each NADH yields about 2.5 ATP and each FADH2 about 1.5 ATP.

Poisons and uncouplers: cyanide blocks Complex IV, so electrons cannot reach oxygen and the whole chain stops. Oligomycin blocks ATP synthase directly. Uncouplers such as 2,4-dinitrophenol (DNP) make the inner membrane leaky to protons, so the gradient collapses and the energy is released as heat instead of ATP. DNP was sold as a diet drug in the 1930s and caused fatal hyperthermia. Brown fat uses this trick on purpose: the protein thermogenin (UCP1) lets protons leak back to generate heat, which is important in newborns and hibernating mammals.

Fermentation and Course Announcements

When oxygen is absent, the electron transport chain stops and NADH cannot be recycled to NAD+. Fermentation solves this by using pyruvate (or a derivative) as the electron acceptor, regenerating NAD+ so glycolysis can keep making its 2 ATP.

Lactic acid fermentation happens in our muscle cells during intense exercise and in the bacteria that make yogurt: pyruvate is reduced directly to lactate. Alcohol fermentation happens in yeast: pyruvate first loses CO2 to become acetaldehyde, which is then reduced to ethanol. The CO2 is what makes bread rise.

Comparison: fermentation yields only 2 ATP per glucose, while full aerobic respiration yields about 30-32, roughly fifteen times more.

Announcements:
- Problem Set 4 is due Friday, October 17 at 11:59 pm on Canvas.
- Midterm 2 covers Lectures 5 through 8 and will be held in the evening on October 28.
- Dr. Okafor's office hours have moved to Thursday 2-4 pm in Hartley Hall 214.
- Next lecture: photosynthesis, which runs much of this pathway in reverse.